
Christopher Barker

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Student developer building frontend, applied AI, data visualization, and interactive systems with portfolio-backed implementation evidence across web apps, Discord bots, research pipelines, and data projects.

SELECTED PROJECTS & EVIDENCE

- [Protein AI Pipeline](#) — Python research pipeline using ESM-3 and BLIP-2 for protein embeddings, image captioning, and similarity evaluation.
- [Pixel Progression](#) — PHP and JavaScript game system with authentication, save states, leaderboards, and gameplay logic.
- [Valorant Discord Bot](#) — Node.js and Discord.js bot using slash commands and the Valorant API for player profiles, rank tracking, match history, and statistics.
- [AI Caption Generator](#) — Streamlit and OpenAI app for image-aware social captions.
- [HTML Resume Portfolio](#) — responsive HTML/CSS portfolio focused on structure, readability, and presentation.
- [Grocery Retail Consumer Analytics](#) — retail analytics using survey data for segmentation and visualization.
- [Opioid Prescribing Risk Analysis](#) — healthcare data analysis using public prescribing data and model interpretation.

EDUCATION

Pacific Lutheran University

February 2024 - May 2026 | Tacoma, WA

Bachelor of Arts, Individualized Major in Web Development and Interactive Design

Tacoma Community College

June 2017 - June 2022 | Tacoma, WA

Associate of Arts, Direct Transfer Agreement

TECHNICAL ACTIVITIES

UX/UI Design Intern / TestUnlimited.com

January 2024 - March 2024

- Converted sketches into UI blueprints and high-fidelity design deliverables.
- Redesigned product page layouts and visual assets to improve usability and clarity.
- Worked with engineering to implement frontend updates that matched approved designs.

Derailed / React, CSS, JavaScript, JSON

June 2025 - September 2025

- Built a browser game interface with React, JavaScript, CSS, and JSON-based state data.
- Designed responsive game controls for browser-based play.
- Implemented player movement, collision logic, and state updates for gameplay interactions.

West Coast Tourism Data Analysis / Python

March 2026

- Analyzed a retail dataset with Python to segment weekend tourism patterns by region, spending tier, and basket composition.
- Produced comparative visualizations surfacing consumer behavior differences across tourism-driven market segments.

AI Protein Sequence Visualization / Python, ESM-3, BLIP-2	June 2025 - Current
- Built a Python research pipeline using ESM-3 and BLIP-2 to generate visual embeddings and natural language captions from protein sequences. - Evaluated embedding-based similarity across protein pairs to support function inference and research interpretation.	

Valorant Discord Bot / Node.js, JavaScript, Discord.js, REST API	May 2026 - June 2026
- Built a Discord slash-command bot that retrieves Valorant player profiles, rank, RR trends, recent match history, and combat statistics. - Integrated external API data and formatted results into readable Discord embeds for competitive performance review. - Added error handling and iterative stat refinement for missing profiles, API responses, and rank-tracking edge cases.	

WORK AND RESEARCH EXPERIENCE

Undergraduate Research Assistant / Pacific Lutheran University	June 2025 - Current Tacoma, WA
- Developed a multimodal AI pipeline combining protein embeddings, image captioning, and similarity analysis for protein function inference. - Applied ESM-3 and BLIP-2 models to extract visual embeddings and natural language captions from protein sequences under faculty supervision. - Contributed to research accepted for presentation at ICIBM 2026 (International Conference on Intelligent Biology and Medicine).	
PC Tech / Pacific Lutheran University	August 2024 - June 2025 Tacoma, WA
Troubleshoot hardware and software issues, supported campus users, and mentored new student workers.	

ACHIEVEMENTS & AWARDS

ICIBM 2026 — Accepted research paper presentation in applied AI and computational biology	2026
Dean's List / University of Washington, Tacoma	June 2024 - Current
First Generation Fellow / University of Washington, Tacoma	June 2023
Outstanding Student Award / Tacoma Community College	2022 - 2023

TECHNICAL SKILLS

Frontend: HTML5, CSS3, JavaScript, PHP, React, Material UI
Applied AI & Data: Python, Streamlit, OpenAI API, ESM-3, BLIP-2, R, data visualization
Programming: Node.js, Discord.js, REST API, Java, JSON, JUnit
Tools: Git, GitHub, Figma, VS Code, Docker, IntelliJ